KEY SHEET

MATHEMATICS

1	**B**	2	**A**	3	**D**	4	**C**	5	**BC**	6	**AD**
7	**AC**	8	**7**	9	**2**	10	**8**	11	**2**	12	**2**
13	**4**	14	**5**	15	**7**	16	**1.57**	17	**1.57**		

PHYSICS

18	**A**	19	**A**	20	**B**	21	**A**	22	**AB**	23	**A**
24	**ABC**	25	**2**	26	**4**	27	**6**	28	**9**	29	**4**
30	**8**	31	**1.66-1.67**	32	**2400**	33	**585**	34	**65**		

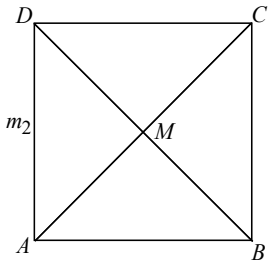
CHEMISTRY

35	**D**	36	**A**	37	**B**	38	**C**	39	**AC**	40	**AB**
41	**A**	42	**0**	43	**2**	44	**5**	45	**8**	46	**5**
47	**5**	48	**5.31-5.37**	49	**2.95-3.10**	50	**116**	51	**206**		

SOLUTIONS

MATHEMATICS

1. $m_1 m_2 = -1 \quad a^2 + 11a + 3\left(m_1^2 + \frac{1}{m_1^2}\right) = 220$



Eq. of AC

$AC = (\cos\alpha - \sin\alpha) + (\sin\alpha + \cos\alpha)y = 10$

$BD = (\sin\alpha - \cos\alpha)x + (\sin\alpha - \cos\alpha)y = 0$

$(10(\cos\alpha - \sin\alpha), 10(\sin\alpha - \cos\alpha))$

Slope of AC $= \left(\dfrac{\sin\alpha - \cos\alpha}{\sin\alpha + \cos\alpha}\right) = \tan\theta = M$

Eq. of line making an angle π_4 with AC

$$m_1, m_2 = \frac{m \pm \tan\dfrac{\pi}{4}}{1 \pm m\tan\dfrac{\pi}{4}} \qquad = \frac{m+1}{1-m} \text{ or } \frac{m-1}{1+m}$$

$$\frac{\dfrac{\sin\alpha - \cos\alpha}{\sin\alpha + \cos\alpha} + 1}{1 - \left(\dfrac{\sin\alpha - \cos\alpha}{\sin\alpha + \cos\alpha}\right)}, \frac{\dfrac{\sin\alpha - \cos\alpha}{\sin\alpha + \cos\alpha} - 1}{1 + \dfrac{\sin\alpha - \cos\alpha}{\sin\alpha + \cos\alpha}}$$

$m_1, m_2 = \tan\alpha, \cot\alpha$

Mid point of AC & BD $= M(5(\cos\alpha - \sin\alpha), 5(\cos\alpha + \sin\alpha))$

$B(10(\cos\alpha - \sin\alpha), 10(\cos\alpha + \sin\alpha))$

$a = AB = \sqrt{2}BM = \sqrt{2}(5\sqrt{2}) = 10$

a = 10 $\qquad \because a^2 + 11a + 3\left(m_1^2 + \dfrac{1}{m_1 2}\right) = 220$

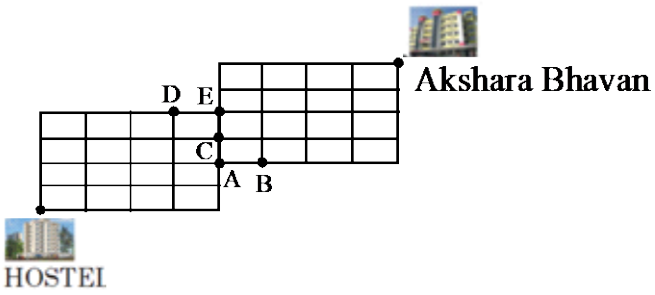
$100 + 110 + 3(\tan^2\alpha + \cot^2\alpha) = 220$

Hence $\tan^2\alpha = 3, \tan^2\alpha = \dfrac{1}{3} \Rightarrow \alpha = \dfrac{\pi}{3}$ or $\dfrac{\pi}{6}$

Now $72\left(\dfrac{9}{16} + \dfrac{1}{16}\right) + 100 - 30 + 13$

$= 72\left(\dfrac{9}{16} + \dfrac{1}{16}\right) + 100 - 30 + 13 \quad = 72\left(\dfrac{5}{8}\right) + 83 = 45 + 83 = 128$

2.



The required paths are

i) Hostel to A to B to AKSHARA BHAVAN

ii) Hostel to C to AKSHARA BHAVAN

iii) Hostel to D to E to AKSHARA BHAVAN

3. $f(x+k+1)=\frac{1}{2}f(x+k)=\frac{1}{2^{k+1}}f(x).$

$$a_n=\int_0^n f(x)dx=\sum_{k=0}^{n-1}\int_k^{k+1} f(x)dx=\sum_{k=0}^{n-1}\int_0^1 f(x+k)dx=$$

$$=\sum_{k=0}^{n-1}\frac{1}{2^k}\int_0^1 f(x)dx\sum_{k=0}^{n-1}\frac{1}{2^k}\to\to\int_0^1 f(x)dx\sum_{k=0}^{\infty}\frac{1}{2^k}=2\int_0^1 f(x)dx, n\to\infty.$$

4. $\mu_{10}=1\mu_{13}=\mu_{16}.....=\mu_{3K+1}$ *for* $K\geq 3$

$\mu_{11}=9=\mu_{14}=\mu_{17}.....=\mu_{3K+2}$

$\mu_{12}=3=\mu_{15}=\mu_{18}.....=\mu_{3K}$

$\mu_{500}=9, \mu_{400}=1, \mu_{300}=3$

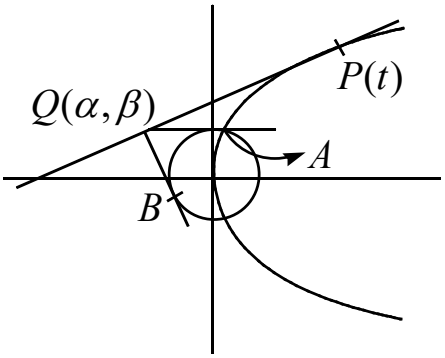
5. B) Select 6 planes out of 11 and arrange A, B, C, D, E, F remaining X, X, X, Y, Y gets arranged in X Y X Y X pattern only at remaining places ${}^{11}P_6$.

C) ${}^{11}C_6\times 1\times\frac{5!}{3!2!}=4620$

6. $P(n)=\sum_{r=0}^{n}(-1)^r\ {}^nC_r-\sum_{r=0}^{n}\frac{(-1)^r\ {}^nC_r}{r+1}$

$$P(n)=\frac{-1}{n+1}\sum_{r=0}^{n}{}^{n+1}C_{r+1}(-1)^r \qquad P(n)=-\frac{1}{n+1}$$

7.



SOL-

Equation of chord of contact

$\alpha x+\beta y=4$ .....(1)

Also, Equation of tangent at P(t)

Is $ty = x + 2t^2$

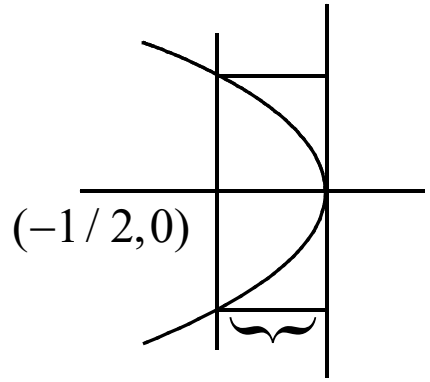
Also $t\beta = \alpha + 2t^2$ .....(2)

From (1) & (2)

$(t\beta - 2t^2)x + \beta y = 4$

$\beta(tx + y) - (2t^2x + 4) = 0 \therefore tx + y = 0 \,\&\, 2t^2x + 4 = 0$

$t = -\dfrac{y}{x} \qquad \therefore 2.\dfrac{y^2}{x^2} \times x + 4 = 0 \; y^2 + 2x = 0$



Area bounded by S = 0 & the line 2x+1=0

$\dfrac{2}{3} \times \dfrac{1}{2} \times 2 = \dfrac{2}{3}$

$2x - y + 1 = 0$ is a focal chord

Hence angle is 90°

8.

x_i	f_i	$f_i x_i$	$f_i x^2{}_i$
A	2	2A	$2A^2$
2A	1	2A	$4A^2$
3A	1	3A	$9A^2$
4A	1	4A	$16A^2$
5A	1	5A	$25A^2$
6A	1	6A	$36A^2$
Total	7	22A	$92A^2$

$\because \sigma^2 = \dfrac{\sum f_i x^2{}_i}{\sum f_i} - \left(\dfrac{\sum f_i x_i}{\sum f_i}\right)^2 \quad \Rightarrow 160 = \dfrac{92A^2}{7} - \left(\dfrac{22A}{7}\right)^2$

$\Rightarrow 160 = \dfrac{92A^2}{7} - \dfrac{484A^2}{49} \Rightarrow 160 = \dfrac{92 \times 7A^2 - 484A^2}{49}$

$\Rightarrow 160 \times 49 = 644A^2 - 484A^2 \Rightarrow 160A^2 = 7840$

$A^2 = \frac{7840}{160} \Rightarrow A^2 = 49 \Rightarrow A = \pm 7$

A = 7 as A is a positive integer

9. Given set is $\{\frac{a}{b} \in I^+ : 2a^2 + 3b^2 = 35, a, b \in Z\}$

We can see that, $2(\pm 2)^2 + 3(\pm 3)^2 = 35$

And $2(\pm 4)^2 + 3(\pm 1)^2 = 35$

$\therefore (2,3),(2,-3),(-2,-3),(-2,3),(4,1),(4,-1),(-4,-1),(-4,1)$

10. $f(m+1) - f(m) = 3(4m-1)$

$f(2) - f(1) = 3(4-1)$

$f(3) - f(2) = 3(8-1)$

$f(19) - f(18) = 3(4.18-1)$

$f(19) = 3[4(9.19-18] = 3.18[37] = 1998$

11. Volume $= \left| \frac{(4-1)(6-3)(12+1)}{\begin{vmatrix} 2 & 2 & 1 \\ 2 & -1 & 1 \\ 3 & 4 & 12 \end{vmatrix}} \right| = \frac{13}{7}$

12. $F(x) = \int_0^x f(t)dt \Rightarrow xF\left(\frac{1}{x}\right) = x\int_0^{1/x} f(t)dt$

Let $\frac{1}{x} = nT + E$

$\lim_{x\to 0} xF\left(\frac{1}{x}\right) = \lim_{x\to\infty} \frac{1}{nT+E} \int_0^{2T+E} f(t)dt$

$= \lim_{x\to\infty} \frac{n}{nT+E} \left\{ \int_0^{nT} f(t)dt + \int_{nt}^{nT+E} f(t)dt \right\}$

$= \lim_{x\to\infty} \frac{n}{nT+E} \int_0^T f(t)dt + \frac{1}{nT+E} \int_0^E f(t)dt \qquad = \frac{1}{T}\int_0^T f(t)dt = \frac{1}{3} \times 6 = 2$

13. $7\cos^2\theta - 3\sin^2\theta - 2\cos^2 2\theta = 2$

$4\cos^2\theta + 3\cos 2\theta - 2\cos^2 2\theta = 2$

$2(1+\cos 2\theta) + 3\cos 2\theta - 2\cos^2 2\theta = 2$

$2\cos^2 2\theta - 5\cos 2\theta = 0$

$\cos 2\theta(2\cos 2\theta - 5) = 0$

$\cos 2\theta = 0 \quad 2\theta = (2n+1)\frac{\pi}{2}$

$\theta = (2n+1)\frac{\pi}{4} \qquad S = \left\{\frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}\right\}$

For all four values of θ

$x^2 - 2(\tan^2\theta + \cot^2\theta)x + 6\sin^2\theta = 0 \Rightarrow x^2 - 4x + 3 = 0$

Sum of roots of all four equations $= 4 \times 4 = 16$.

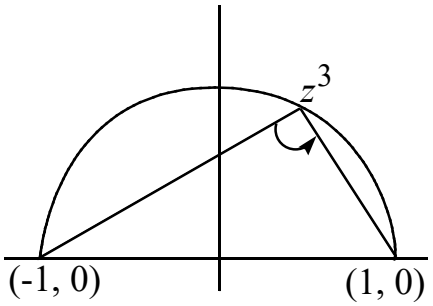
14&15. *Let* $\cot A = a$ *then* $a^3 + a^2|a + x| + |a^2x + 1| = 1$

$|a^3 + a^2x| + |a^2x + 1| = (a^2x + 1) - (a^2x + a^3)$

Finally we get, $a \in (-\infty, -5] \cup \left[-\frac{1}{\sqrt{5}}, \frac{1}{\sqrt{6}}\right]$

$\mathrm{Re}(z) > 0$ is ___

16.&17. Sol- w lie on $11z^{10} + 10iz^9 + 10iz - 11 = 0 \Rightarrow |w| = 1$



$\arg\left(\frac{z^3 - 1}{z^3 + 1}\right) = \frac{\pi}{2}$

Locus of z^3 will be in I, II quadrant and lie on a unit circle

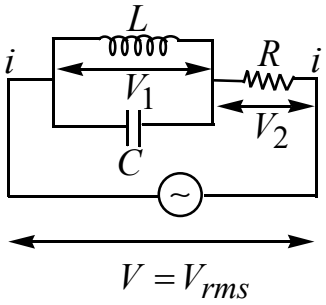
PHYSICS

18. APPLY BERNOILLLIS EQUATION

19. CONCEPTUAL

20. $V_1^2 + V_2^2 = V^2 \qquad \frac{V_1}{X_L} - \frac{V_1}{X_C} = i$

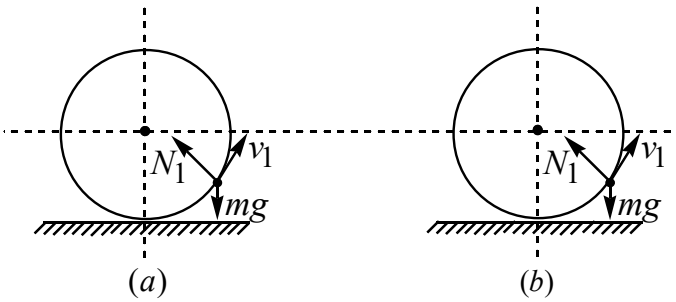
$$V_1\left(\frac{1}{\omega L} - \omega C\right) = i \quad V_1 = \frac{\omega L i}{1-\omega^2 LC} \qquad V_2 = iR$$



$V = V_{rms}$

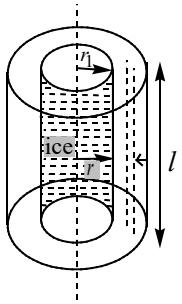
$$\frac{\omega^2 L^2}{\left(1-\omega^2 LC\right)^2} + R^2 = \frac{V^2}{i^2} \quad i = \frac{V}{\sqrt{R^2 + \frac{\omega^2 L^2}{\left(1-\omega^2 LC\right)^2}}}$$

21.



F.B.D in two cases is as shown in case (b) normal will be less and incase (b) work done by magnetic field is zero so height in two cases will be same.

23.



$$R_{eq} = \int_{r_1}^{r_2} \frac{dr}{k.2\pi r\ell} = \frac{\ell n 2}{k.2\pi\ell}$$

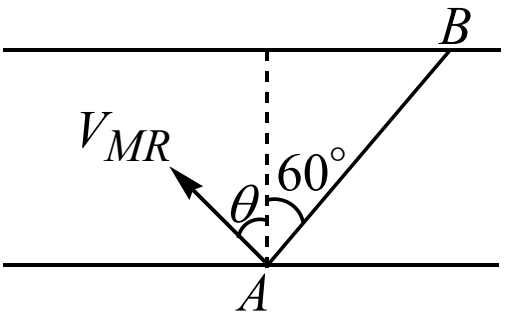
$$\therefore \text{ Heat current} = \frac{T_s - T_{ice}}{R_{eq}} = \frac{40}{\ell n 2} k 2\pi\ell = 80\pi\, J/\sec$$

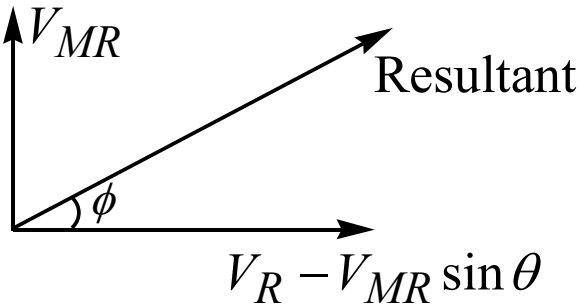
Also, $\frac{dm_{ice}}{dt}L_f = 80\pi \quad \Rightarrow \frac{dm_{ice}}{dt} = \frac{80\pi}{80\times 4.2\times 1000} = \frac{\pi}{4200} kg/\sec$

23. CONCEPTUAL

24. CONCEPTUAL

25.





SOL-

$\tan\phi = \frac{V_{MR}\cos\theta}{V_R - V_{MR}\sin\theta}; \phi = 30^\circ$

On solving $\frac{V_R}{\sqrt{3}\cos\theta + \sin\theta} = V_{MR}$

V_{MR} will be minimum when denominator is maximum. That is at $\theta = 30^\circ$.

And time $= \frac{100\sqrt{3}}{V_{MR}\cos\theta} = 200\sqrt{2}$

26. Since BC is shorted, potential difference across BC is zero.

Voltmeter reading $= V_{AB} + V_{BC} = V_{AB} = 8$ Volt (given)

Since length CD $= 2\times$ length AB, $V_{CD} = 2\times V_{AB} = 16$ Volt

Therefore emf of cell $= V_{AB} + V_{BC} + V_{CD} = 8 + 0 + 16 = 24$ Volt

On removing the resistance less wire connecting B and C.

$V_{AC} = 2V_{CD}$

$V_{AC} = V_{CD} = 24$

$V_{AC} = 16$

27. u = 5.0 cm

v = 10.0 cm

$f = \frac{uv}{u+v} = \frac{10}{3} cm$

$$\frac{df}{f} = f\left(\frac{du}{u^2} + \frac{dv}{v^2}\right)$$

$$\frac{df}{f} = f(du)\left(\frac{1}{u^2} + \frac{1}{v^2}\right)$$

% error in focal length

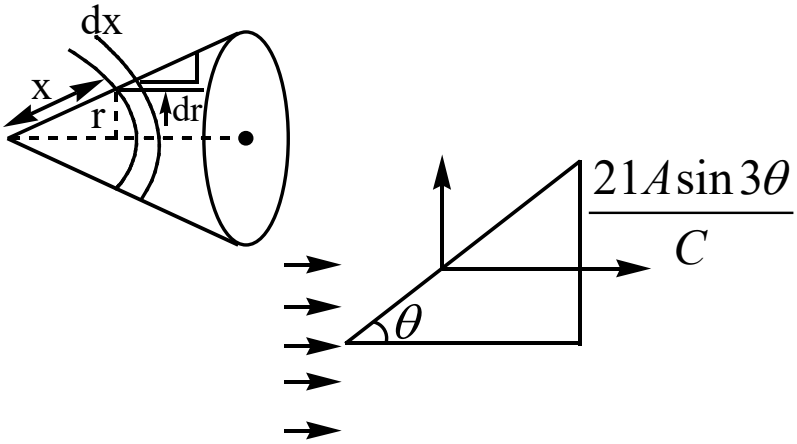
$$= f(du)\left(\frac{1}{u^2} + \frac{1}{v^2}\right)100 = \frac{5}{3}$$

28. $$dF_R = \frac{2I_1.dA.\sin^2\theta}{C}$$

$$\Rightarrow \frac{2.I_0 r^3}{R^3}.\frac{2\pi r.dx.\sin^2\theta.\sin\theta}{C}$$

$$\Rightarrow \frac{4\pi I_0 r^4 dr\sin^2\theta}{R^3.C}$$

$$F_R = \frac{4\pi I_0.R^2.\sin^2\theta}{5C}$$



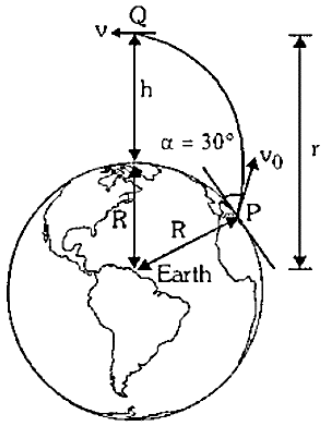
29. Conceptual

30. Let the velocity at highest point be v and R + h=r

Applying conservation of angular momentum between P and Q, we have

$$mvr = mv_0 R\cos 30^\circ$$

Or $$v = \frac{\sqrt{3}v_0 R}{2r}$$



Applying conservation of mechanical energy between P and Q, we have:

$$\frac{1}{2}mv_0^2 - \frac{GMm}{R} = \frac{1}{2}mv^2 - \frac{GMm}{r}$$

Substituting the value of v from equation (1), we get

$$\frac{1}{2}mv_0^2 - \frac{GMm}{R} = \frac{1}{2}m\left\{\frac{3v_0^2R^2}{4r^2}\right\} - \frac{GMm}{r}$$

Or $v_0^2 - \frac{2GM}{R} = \frac{3v_0^2R^2}{4r^2} - \frac{2GM}{r}$

Or $\frac{1.5GM}{R} - \frac{2GM}{R} - \frac{3}{4}\left(\frac{1.5GM}{R}\right)\frac{R^2}{r^2} - \frac{2GM}{r}$

$\left(v_0 = \sqrt{\frac{1.5GM}{R}} \text{ is given}\right)$

Or $-\frac{1}{2R} = \frac{9}{8R}.\frac{R^2}{r^2} - \frac{2}{r}$

Or $-4r^2 = 9R^2 - 16Rt$

Or $4r^2 - 16Rr + 9R^2 = 0$

Or $r = \frac{16R + \sqrt{256R^2 - 144R^2}}{8}$

$\Rightarrow = \frac{16R \pm 10.58R}{8} = 3.323R \text{ or } 0.677R$

But r < R

Hence r = 3.323 R

Or h = r – R = 2.323 R

31. $f = \frac{C - V_R}{C + V_R} f_0$

$\frac{\Delta f}{f_0} = \frac{2V_R}{C + V_R} \approx \frac{2V_R}{C} \qquad \Rightarrow C = \frac{2 \times 0.1}{600} \times 5 \times 10^6 = 1666.66 m/s$

32. $f' = \frac{C - V_R}{C + V_R} f_0 \qquad \frac{\Delta f}{f_0} = \frac{2V_R}{C + V_R}$

Velocity is 4 times $\Rightarrow \Delta f = 4\Delta f_0 = 2400\,Hz$

33.& 34.

P_0, V_0, T_0 A_1	P_0, V_0, T_0 A_2	P_0, V_0, T_0 A_3

Initial condition

P, V_1, T_1 A_1	$P, V, \frac{9T_0}{4}$ A_2	$P, V, \frac{9T_0}{4}$ A_3

final equilibrium

Considering chamber A_2 and A_3 ,

No. of moles = constant $\qquad \frac{P_0 2V_0}{2RT_0} = \frac{P2V}{2\frac{9T_0}{4}}$

$$PV = \frac{9}{4} P_0 V_0$$

Apply adiabatic condition

$P_0(2V_0)^\gamma = P(2V)^\gamma \qquad V = \left|\frac{2}{3}\right|^3 V_0$

$P_0(V_0)^\gamma \; 2^\gamma = P 2^\gamma v^\gamma \qquad V = \frac{8}{27} V_0$

$P_0 V_0^\gamma = PV^\gamma$

$V^\gamma = \frac{P_0 V_0^\gamma}{P} \qquad V_1 = 3V_0 - 2V = 3V_0 - \frac{16V_0}{27}$

$V^\gamma = \frac{P_0 V_0^\gamma}{\frac{9}{4}\frac{P_0 V_0}{V}} \qquad = \frac{81V_0 - 16V_0}{27}$

$V^{\gamma-1} = \frac{4}{9} V_0^{\gamma-1} \qquad V_1 = \frac{65V_0}{27}$

$V^{2/3} = \frac{4}{9} V_0^{2/3}$ From above volume we can calculate the final pressure.

$V^2 = \left(\frac{4}{9}\right)^3 V_0^2$ Final pressure $P = \frac{243}{32} P_0$ and

$V = \frac{4}{9}^{3/2} V_0$ Final temperature $T = \frac{585}{32} T_0$

CHEMISTRY

35. $XeF_4 + O_2F_2 \xrightarrow{143K} XeF_6 + O_2$

36. $CO_2\,(115\,pm)$

Normal C=O bond length is: 121pm

37. $K_W = 10^{-13}$

$d = 0.9\ gm/ml$

1000 ml water = 900 gram water

$$\text{Molarity} = \frac{900 \times 1000}{18 \times 1000}$$

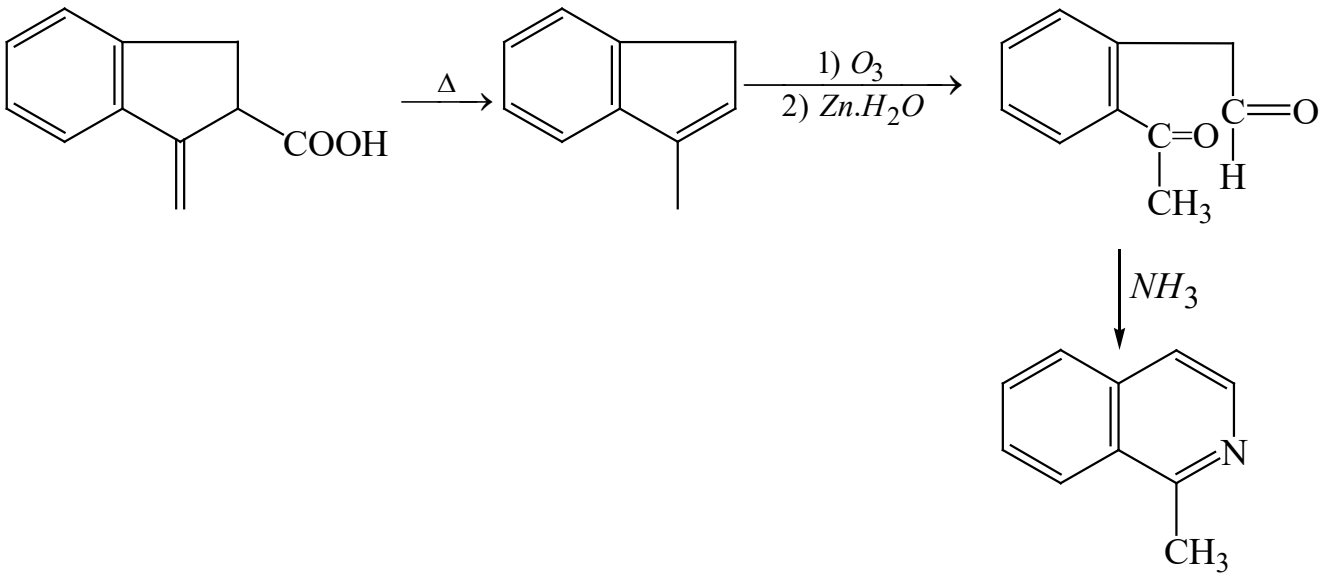
$= 50$

$$\underset{C(1-\alpha)}{H_2O} \rightleftharpoons \underset{C\alpha}{H^+} + \underset{C\alpha}{OH^-}\,[K_a]$$

$$K_a = \frac{[H^+][OH^-]}{[H_2O]} = \frac{10^{-13}}{50}$$

$= 2 \times 10^{-15}$

38.

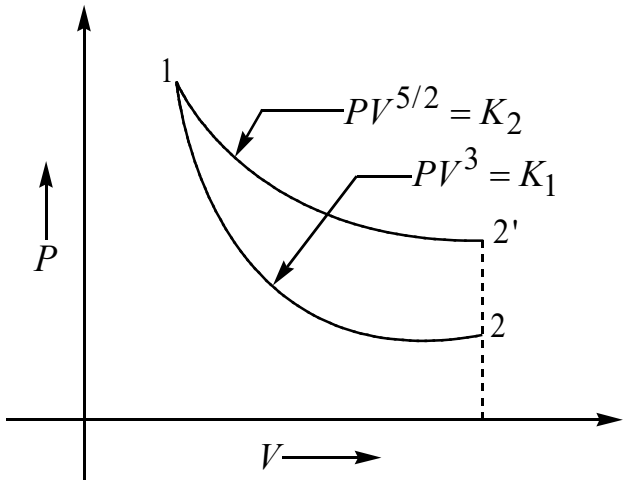


39. $Cu^{2+}, Pb^{2+}, Hg^{2+}, Bi^{3+}$:*IIA group*

Zn^{2+}, Co^{2+} : *IV group*

40.

41.



$1 \rightarrow$ Initial state
$2 \rightarrow$ Final state of the 1st process
$2' \rightarrow$ Final state of the IInd process

Work done in 2nd process is more than 1st process.

42. $X = \left[Ni(CO)_4 \right]$ and $Y = ZrI_4$

43. $\boxed{\overset{+4}{ClO_2} + NaOH \rightarrow \overset{+3}{NaClO_2} + \overset{+5}{NaClO_3}}$

44.

45. Conceptual

46. amount of Ag deposited $= (675 \times 0.0004) \times 10$ *gram*

$$\frac{w}{E} = \frac{i \times t}{96500}$$

$$\frac{(675 \times 0.0004) \times 10}{108} = \frac{96.5 \times t}{2 \times 96500} [Eq.wt\ of\ Ag = 108]$$

t = 50 sec

$$\Rightarrow \frac{x}{10}$$

$$\Rightarrow \frac{50}{10} = 5$$

47. For 1st line of Balmer

Series of He^+

$$\Delta E = E_3 - E_2$$

$$= \frac{-13.6 \times (2)^2}{(3)^2} + \frac{13.6 \times (2)^2}{(2)^2}$$

$$= 7.55\ eV$$

For Li^{+2}

$$E_5 = \frac{-13.6 \times (3)^2}{(5)^2}$$

$$= -4.896\ eV$$

$$E_4 = \frac{-13.6 \times (3)^2}{(4)^2} = -7.65\ eV$$

So electron is emitted out from $n = 5$

48 & 49

$\therefore$ rate constant at $310K = k + \frac{7k}{100} = 1.07k$

Using, $2.303\log\frac{k_2}{k_1} = \frac{E_a^f}{R}\left[\frac{T_2 - T_1}{T_1 T_2}\right]$

$$2.303\log\frac{1.07k}{k} = \frac{E_a^f}{8.314}\left[\frac{310-300}{310\times 300}\right]$$

$E_a^f = 5.34kJ$

Now equilibrium constant at $300K = K_c$

$\therefore$ equilibrium constant at $310K = K_c + \frac{3K_c}{100} = 1.03K_c$

Using, $2.303\log\frac{K_2}{K_1} = \frac{\Delta H}{R}\left[\frac{T_2 - T_1}{T_1}\right]$

$$2.303\log\frac{1.03K_c}{K_c} = \frac{\Delta H}{8.134}\left[\frac{310-300}{310\times 300}\right]$$

$$\Delta H = 2.31kJ$$

Since, $\Delta H = E_a^f - E_a^b$

$E_a^b = 3.03kJ$

50 & 51

